

ClayQuant

Calculation of X-ray powder diffraction patterns of interstratified clay minerals, and quantification of clay mineral assemblages from oriented mounts

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Scope. ClayQuant computes one-dimensional ($00l$) and three-dimensional (hkl) X-ray powder diffraction patterns for mica-type (illite), chlorite and kaolinite layer structures, for the ethylene glycol–smectite complex, and for randomly or non-randomly interstratified two-component stacks of these layers, with arbitrary preferred orientation and crystallite thickness distribution. It then uses those calculated patterns to quantify the clay mineral assemblage of a clay separate measured as three oriented mounts (air-dried, ethylene glycol-solvated and heated), by non-negative least squares over the whole pattern.

The program is written in Python, is operated through a local browser interface or from the command line, and is distributed as source. This manual documents the physical model, the implementation, the operating procedure, every control of the interface, the output files, and the tests by which the program was verified.

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1 Introduction

1.1 Purpose

Quantitative analysis of clay mineral assemblages by X-ray diffraction is made difficult by three properties of the minerals themselves. Clay minerals are platy, so a specimen prepared for analysis is strongly textured rather than randomly oriented. They are frequently interstratified, so that a crystallite is a stack of two or more different layer types and produces neither the pattern of one end member nor a superposition of both. And they are finely divided, so that crystallites are only a few tens of layers thick and their reflections are correspondingly broad. None of these is a defect of the measurement; all three are properties of the material that a calculation must reproduce before a quantification can mean anything.

ClayQuant addresses these by calculating the pattern that each candidate phase would produce *under the conditions of the measurement* — its texture, its crystallite thickness, its layer spacing, the instrument's emission profile and slit geometry — and then fitting the measured pattern as a non-negative combination of those calculated patterns.

1.2 What the program does and does not do

It does:

- compute $00l$ and hkl intensities from crystal structures supplied as CIF;
- compute the diffracted intensity of a two-component interstratified stack with arbitrary junction probabilities and a distribution of crystallite thicknesses;
- apply preferred orientation, Lorentz–polarization, beam overflow, an arbitrary emission profile and an adjustable peak shape;
- read the common vendor file formats, determine the 2θ zero error, model the background, identify kaolinite and expandable clay from the three-mount comparison, identify accompanying non-clay minerals, and fit and report the assemblage.

It does not:

- refine structures. Cell parameters, atomic coordinates, site occupancies and displacement parameters are taken as given and are not adjusted against the data. ClayQuant is not a Rietveld program;
- refine peak positions, widths or orientation continuously. The fit chooses among *pre-computed* patterns, so the achievable agreement is limited by how finely the library samples those parameters (§3.3, §9);
- report weight percent. What is reported is each phase's share of diffracted intensity; the conversion requires a reference intensity ratio per phase (§2.13);
- model three-component interstratifications, or interstratifications with a lateral (ab) component of disorder beyond that implied by the treatment of §2.12.

1.3 Notation and conventions

Throughout, λ is the wavelength in ångström, θ the Bragg angle, and

$$s = 2 \sin\theta / \lambda = 1/d \quad (1.1)$$

the length of the scattering vector, in $\text{\AA}^{-1}$. Note that s is defined here without the factor 2π and without the factor $1/2$ used in some crystallographic literature; the quantity $q = \sin\theta/\lambda = s/2$ is used only as the argument of scattering factors and displacement factors, matching International Tables convention. The stacking axis of a layer structure is c^* , and heights along it are given in ångström and denoted z . Layer repeat distances are denoted d_A , d_B . Intensities are per unit incident flux and are reported in arbitrary units, since only relative intensities enter the analysis.

1.4 Related programs

The one-dimensional part of the calculation follows the same physics as NEWMOD (Reynolds), PyXRD (Dumon & Van Ranst) and Sybilla, all of which implement the matrix formalism of §2.5. ClayQuant differs from these chiefly in coupling the calculation directly to a whole-pattern non-negative least-squares quantification, in computing the three-dimensional hkl contribution of the host phases from crystal structures rather than treating only basal reflections, and in being open source and scriptable.

2 Theory

2.1 The oriented mount

A clay separate is deposited on a glass slide so that the platelets lie with (001) close to the slide surface. In Bragg–Brentano reflection geometry the diffraction vector is always parallel to the specimen normal, so such a specimen presents predominantly its basal (00 l) reflections. The degree to which this is true is a property of the preparation, not a constant, and is treated here as an adjustable texture parameter (§2.9).

Three mounts of the same separate are measured:

Mount	Treatment	Diagnostic role
Air-dried	as deposited, equilibrated with laboratory air	reference state
Glycol-solvated	exposed to ethylene glycol vapour	expandable layers swell to a two-layer complex near 16.9 $\text{\AA}$
Heated	typically 500 °C	kaolinite dehydroxylates and its 7.15 $\text{\AA}$ reflection is lost; chlorite survives

The quantitative fit is performed on the glycol-solvated mount, because glycolation separates the expandable from the non-expandable component and resolves the illite/smectite series, which in the air-dried state overlap near 10–14 $\text{\AA}$.

2.2 Scattering factors and the structure factor

Atomic scattering factors $f_0(q)$ are evaluated from the eleven-parameter analytical expansion of Waasmaier & Kirfel (1995),

$$f_0(q) = c + \sum_{i=1}^5 a_i \exp(-b_i q^2) \quad (2.1)$$

valid to $q = 6 \text{ \AA}^{-1}$, with $q = \sin\theta/\lambda$. Ionic species are used where the tabulation provides them and the CIF specifies a charge; otherwise the neutral atom is used, and a species that is not tabulated falls back to its neutral atom with a warning. Isotropic displacement is applied as

$$T(q) = \exp(-B q^2) \quad (2.2)$$

with B in $\text{\AA}^2$. Where a CIF reports U_{iso} , the conversion $B = 8\pi^2 U$ is applied. The structure factor of a reflection hkl is then

$$F(hkl) = \sum_j n_j f_j(q) \exp(-B_j q^2) \exp[2\pi i(hx_j + ky_j + lz_j)] \quad (2.3)$$

summed over the symmetry-expanded content of the unit cell, with n_j the site occupancy. Symmetry operations are taken from the CIF as written, parsed into a rotation and a translation, applied to each asymmetric-unit site, and duplicate positions removed within a tolerance of 10^{-4} in fractional coordinates.

2.3 The one-dimensional layer model

For a basal reflection $h = k = 0$, equation (2.3) reduces to

$$F(00l) = \sum_j n_j f_j T_j \exp(2\pi i l z_j) \quad (2.4)$$

which depends only on the fractional coordinate along c , whatever the cell angles: the scalar product of the reciprocal vector $(00l)$ with a position $x\mathbf{a} + y\mathbf{b} + z\mathbf{c}$ is lz exactly. No projection correction is required. The basal spacing follows from the triclinic metric as

$$d(001) = 1/|\mathbf{c}^*| = V / (ab \sin\gamma) \quad (2.5)$$

A *layer model* is therefore a list of scatterers at absolute heights z (in ångström) within one layer of repeat distance d , each with an occupancy and a displacement parameter, together with the continuous layer structure factor

$$F(s) = \sum_j n_j f_j(s/2) \exp(-B_j s^2/4) \exp(2\pi i s z_j) \quad (2.6)$$

Here s is continuous, not restricted to l/d : an interstratified stack is not periodic and scatters at every s , which is precisely what §2.5 integrates.

Deriving a layer model from a polytype cell

The structures available for illite, chlorite and kaolinite are multi-layer polytypes: illite ICSD 90144 is a two-layer $C2/c$ cell, kaolinite $2M$ ICSD 30285 a two-layer $C1c1$ cell. Successive layers in these polytypes are related by operations of the form $(x, -y, z+\frac{1}{2})$ — a rotation or reflection in the ab plane combined with a translation of exactly half the cell along c . Such an operation leaves the projection of the electron density on c^* unchanged. The projected density therefore repeats every $d(001)/n$, and a single-layer model is obtained exactly by folding all atomic heights modulo that distance. This is done automatically; the fold is exact for the $2M_1$ and $I1b$ polytypes used here and was verified numerically (§A.1).

Consequence. Because only the projection on c^* enters, kaolinite $1M$ and kaolinite $2M$ have nearly identical basal series and are not distinguishable from $00l$ intensities alone. They are separable only through their hkl reflections, which in a strongly oriented mount are weak. Both are provided, and a fit that divides intensity between them should not be read as a polytype determination.

2.4 Basal reflections of a periodic stack

For a crystallite of N identical layers of thickness d , the amplitudes add with a phase $2\pi s d$ per layer and

$$I(s) = |F(s)|^2 \cdot \sin^2(\pi N s d) / \sin^2(\pi s d) \quad (2.7)$$

the familiar Laue interference function, which has maxima of height N^2 at $s = l/d$ and a width in s of about $1/(Nd)$. Equation (2.7) is the $A = B$ limit of the general result below, and is used as a closed-form check on it (§A.1).

2.5 Interstratification: the matrix method

Consider a crystallite of N layers stacked along c^* , each of type A or B, with layer structure factors $F_A(s)$, $F_B(s)$ and thicknesses d_A , d_B . Let the sequence be a stationary first-order Markov chain with transition matrix $\mathbf{P}$, where P_{mn} is the probability that a layer of type m is followed by one of type n , and with stationary proportions $\mathbf{W}$ satisfying $\mathbf{W}\mathbf{P} = \mathbf{W}$.

The scattered amplitude is

$$A(s) = \sum_{j=1}^N F_{t_j}(s) \exp(2\pi i s z_j), \quad z_j = \sum_{k < j} d_{t_k} \quad (2.8)$$

and the expected intensity $\langle |A|^2 \rangle$ separates into a self term and a pair term,

$$I(s) = \sum_j \langle |F_{t_j}|^2 \rangle + 2 \operatorname{Re} \sum_{j < k} \langle F_{t_k} F_{t_j}^* \exp(2\pi i s (z_k - z_j)) \rangle \quad (2.9)$$

Define the matrix

$$T_{mn}(s) = P_{mn} \exp(2\pi i s d_m) \quad (2.10)$$

Then the sum over all sequences of n steps from type m to type p , weighted by probability and carrying the accumulated phase, is exactly $[\mathbf{T}^n]_{mp}$. Writing $f_m = W_m F_m^*$, the pair at separation n contributes $\mathbf{f}^T \mathbf{T}^n \mathbf{F}$, and counting the $(N-n)$ pairs at that separation gives

$$I(s) = N \sum_m W_m |F_m|^2 + 2 \operatorname{Re} \sum_{n=1}^{N-1} (N-n) \mathbf{f}^T \mathbf{T}^n \mathbf{F} \quad (2.11)$$

which is the result implemented. Dividing by N and letting $N \rightarrow \infty$ gives the Kakinoki–Komura form

$$I_\infty(s)/N = \operatorname{Re} \{ \mathbf{f}^T [2(\mathbf{I}-\mathbf{T})^{-1} - \mathbf{I}] \mathbf{F} \} \quad (2.12)$$

since $\sum_{n \geq 1} \mathbf{T}^n = (\mathbf{I}-\mathbf{T})^{-1} - \mathbf{I}$ and $\mathbf{f}^T \mathbf{F} = \sum W_m |F_m|^2$. ClayQuant uses the finite form (2.11), because crystallite thickness controls basal peak width and the infinite limit would give delta functions.

On the phase convention. The distance from the origin of layer j to that of layer $j+1$ is the thickness of layer j , not of layer $j+1$. This is why the exponential in (2.10) carries d_m , the thickness of the layer being *left*. Using the other convention transposes the roles of the two components and is a common source of error; the implementation is checked against the pure end members and against the closed form (2.7) for this reason (§A.1).

2.6 Crystallite thickness

Real specimens contain crystallites of many thicknesses. Averaging (2.11) over a distribution $p(N)$ and normalising per layer gives

$$I(s)/\langle N \rangle = \sum_m W_m |F_m|^2 + 2 \operatorname{Re} \sum_{n \geq 1} G(n) \mathbf{f}^T \mathbf{T}^n \mathbf{F} \quad (2.13)$$

with the pair weights

$$G(n) = \langle \max(N-n, 0) \rangle / \langle N \rangle \quad (2.14)$$

so the thickness distribution enters through one vector of weights, computed once and reused at every s . The distribution used is lognormal in N , parameterised by its mean $\langle N \rangle$ and by the width β of the underlying normal distribution in $\ln N$, following the crystallite size distribution (CSDS) treatment standard in this field (Drits, Eberl & Środoń). $G(n)$ is normalised so that $G(0) = 1$.

Crystallite thickness is the dominant control on basal peak width. For a 10 Å layer, a mean of 10 layers gives a 001 width of order $0.8^\circ 2\theta$, while a well-crystallised illite may show 0.12° , which requires of order 70 layers (§A.7). It cannot be assumed; it must either be spanned by the library or determined from the data.

2.7 Ordering and Reichweite

The transition matrix carries the ordering. For a two-component system with proportions W_A , W_B :

Case	Transition probabilities	Description
Random, $R = 0$	$P_{AB} = W_B, P_{BA} = W_A$	a layer's identity is independent of its neighbour
Ordered, $R = 1$	$P_{AB} > W_B$	unlike layers tend to alternate; the limit $P_{AB} = 1$ at $W_A = \frac{1}{2}$ is a regular alternation (rectorite-like)
Segregated	$P_{AB} < W_B$	like layers cluster; the limit $P_{AB} = 0$ is a physical mixture

Ordering is expressed through a single parameter P_{AB} , from which the remaining probabilities follow by the stationarity and normalisation conditions $W_A P_{AB} = W_B P_{BA}$ and $P_{mA} + P_{mB} = 1$. Values of P_{AB} for which any probability would fall outside $[0,1]$ are rejected. The supplied reference library is computed for the random case, which is the appropriate default for illite/smectite and chlorite/smectite in most sedimentary settings; ordered stacks can be computed in the simulator or by calling the library builder directly.

2.8 The Lorentz-polarization factor

Which Lorentz factor applies to an oriented clay aggregate measured on a diffractometer is a genuine and long-standing question, and the choice matters: the two candidate conventions differ by a factor l across a basal series after normalisation at 001, so an incorrect choice distorts every relative basal intensity and therefore every quantification derived from them.

The two candidates are the random-powder (Debye–Scherrer) form

$$LP_{\text{powder}} = (1 + \cos^2 2\theta) / (2 \sin^2 \theta \cos \theta) \quad (2.15)$$

and the single-crystal form appropriate to a one-dimensionally periodic, perfectly oriented specimen,

$$LP_{\text{crystal}} = (1 + \cos^2 2\theta) / (2 \sin 2\theta) \quad (2.16)$$

Bradley (1954) and Schoen (1962) found the random-powder factor to work for oriented aggregates; MacEwan, Amil & Brown (1961) argued for something intermediate. The matter was settled empirically by Reynolds (1965), who showed for an ethylene glycol–montmorillonite complex (his Table 2) and independently for a dry Na-montmorillonite (his Table 4) that the random-powder factor is *required* to reproduce observed basal intensities, the single-crystal factor giving large deviations — for the dry montmorillonite, a calculated 003 intensity of 141 against an observed 55.

ClayQuant therefore uses (2.15) by default. This is not merely a convention: the published one-dimensional clay layer models, including the glycol–smectite model used here, were refined under it, and their structure factors are only consistent when it is applied. Equation (2.16) is selectable for comparison. A monochromator-modified polarization factor,

$$p = (1 + \cos^2 2\theta \cos^2 2\theta_M) / (1 + \cos^2 2\theta_M) \quad (2.17)$$

is available for a crystal-monochromated beam.

2.9 Preferred orientation

Texture is described by the March–Dollase model (March 1932; Dollase 1986). For a reflection whose scattering vector makes an angle α with the texture axis,

$$P(\alpha; r) = (r^2 \cos^2 \alpha + \sin^2 \alpha / r)^{-3/2} \quad (2.18)$$

The texture axis is c^* , the platelet normal, which in Bragg–Brentano reflection geometry coincides with the specimen normal. $r = 1$ is a random powder; $r < 1$ describes platelets lying in the specimen plane, enhancing $00l$ and suppressing hkl . The function conserves total scattered intensity, its average over the sphere being unity, so r redistributes intensity rather than creating it.

Preferred orientation does not change a purely basal pattern. For every $00l$ reflection $\alpha = 0$, so $P = r^{-3}$ for all of them alike: a common scale factor. The March–Dollase parameter therefore has no effect on the *shape* of a basal series, and acts only on the ratio of basal to non-basal intensity. This is why the ten orientation parameters of the reference library change an interstratified pattern only through its hkl contribution (§2.12), and why r cannot be determined from basal reflections alone.

2.10 Beam overflow

The Lorentz factor diverges as $1/\sin^2 \theta$, but a real measurement does not, because at low angle the irradiated length of a flat specimen,

$$L_{\text{irr}}(\theta) = R \gamma / \sin \theta \quad (2.19)$$

with R the goniometer radius and γ the equatorial divergence in radians, exceeds the specimen and part of the beam misses it. Only the intercepted fraction diffracts:

$$\varphi(\theta) = \min[1, L_{\text{spec}} \sin \theta / (R \gamma)] \quad (2.20)$$

This matters precisely where clay basal reflections lie. Reynolds (1965) avoided the correction by changing beam slits with angle so that the beam never exceeded his specimen, following Klug & Alexander (1954); with the fixed slits of a modern instrument it must be applied, or every calculated pattern carries an unphysical low-angle ramp. The specimen length, goniometer radius and divergence are settable; for a variable-divergence (automatic) slit, which keeps the irradiated length constant, the correction should be disabled.

2.11 Emission profile and peak shape

The source is described as a sum of Lorentzian-weighted lines, each with its own wavelength, relative intensity and intrinsic width. The default Cu K α profile is the five-line description

Line	λ (Å)	Relative weight	Width
1	1.534753	0.0159	3.6854
2	1.540596	0.5691	0.4370
3	1.541058	0.0762	0.6000
4	1.544410	0.2517	0.5200
5	1.544721	0.0871	0.6200

Every reflection is generated once per emission line, at the 2θ corresponding to that line's wavelength and scaled by its weight, so the $K\alpha_1/K\alpha_2$ separation emerges naturally and correctly with angle rather than being imposed. Other profiles, including a single $K\alpha_1$ line, are available.

The instrumental profile is a pseudo-Voigt whose Gaussian width follows the Caglioti relation

$$\text{FWHM}^2 = U \tan^2\theta + V \tan\theta + W \quad (2.21)$$

with a mixing parameter η between Gaussian and Lorentzian, optionally combined with a Scherrer broadening from a finite in-plane crystallite dimension. Note that broadening along c^* is *not* imposed through (2.21): for interstratified phases it arises physically from the crystallite thickness distribution of §2.6.

2.12 Assembling a pattern

A pattern of a discrete crystalline phase is computed by enumerating every reciprocal lattice point within the measurable range, evaluating (2.3), and applying preferred orientation, Lorentz–polarization and beam overflow before convolution with the emission profile and peak shape. Reflections are enumerated individually rather than by symmetry-equivalent family, so multiplicity is accounted for by the enumeration itself and each reflection receives its own March–Dollase factor according to its own angle to c^* — which is what the texture correction requires.

An interstratified pattern is assembled from two contributions:

1. the continuous basal intensity of §2.5, evaluated at every point of the 2θ grid and carrying LP, beam overflow and the emission profile;
2. the non-basal (hkl , $k \neq 0$ or $h \neq 0$) reflections of the host phase, computed from its crystal structure and scaled by their March–Dollase factors.

Approximation. Contribution (2) treats the non-basal scattering of an interstratified crystal as that of its host layer type, scaled by the host's proportion. This is an approximation. In reality, turbostratic stacking produces two-dimensional hk bands with a characteristic sawtooth profile rather than sharp three-dimensional reflections, and the interstratified component contributes its own non-basal scattering. The approximation is acceptable for a strongly oriented mount, where non-basal intensity is heavily suppressed by the texture, and it degrades as r approaches 1. The relative scaling of the two contributions

is fixed by computing, for a pure host stack, the ratio of its basal intensity in the two formulations, so that a pure end member reproduces its own three-dimensional calculation (§A.1).

2.13 Quantification

The measured pattern $y(2\theta)$, after zero-error correction and background subtraction, is fitted as

$$y(2\theta) \approx \sum_p c_p y_p(2\theta), \quad c_p \geq 0 \quad (2.22)$$

over the library patterns y_p , by non-negative least squares. The non-negativity constraint is physical — a phase cannot be present in negative amount — and has the useful side effect of producing a sparse solution, so that of several hundred candidate patterns only a handful take non-zero coefficients.

The solve is weighted. The variance of a counting measurement at a point is the number of counts recorded there, so the weight is

$$w_i = 1 / \max(y_{\text{raw},i}, 1) \quad (2.23)$$

taken from the *raw* counts, before background subtraction: subtracting a background does not reduce the Poisson variance of the measurement. Weighting instead by the background-subtracted intensity — an easy error — places the greatest weight on the flat regions between peaks, where the subtracted value is near zero and only noise remains, and almost none on the peaks that carry the information (§A.9). Agreement is reported as

$$R_{\text{wp}} = [\sum w_i (y_i - y_{\text{calc},i})^2 / \sum w_i y_i^2]^{1/2}, \quad R_p = \sum |y_i - y_{\text{calc},i}| / \sum |y_i| \quad (2.24)$$

From coefficients to proportions

Two measures are reported for each phase, and they answer different questions:

Scattering share

the phase's fraction of the total calculated integrated intensity, $c_p y_p / \sum c_q y_q$. This is the quantity the diffractogram actually measures.

Amplitude share

the phase's fraction of the summed scale factors, $c_p / \sum c_q$. Broad interstratified patterns carry far more integrated area per unit peak height than a sharp discrete phase, so the two can differ by a factor of several.

Neither is a weight percent. Converting a scattering share to a mass fraction requires a reference intensity ratio per phase — the calculated intensity per unit mass in the same geometry — and for an oriented mount it depends additionally on how the mount was prepared, since texture differs between phases and between preparations. A method is provided to apply calibrated ratios if they are available (`Quantification.weight_percent`); in their absence the numbers are relative and should be reported as such.

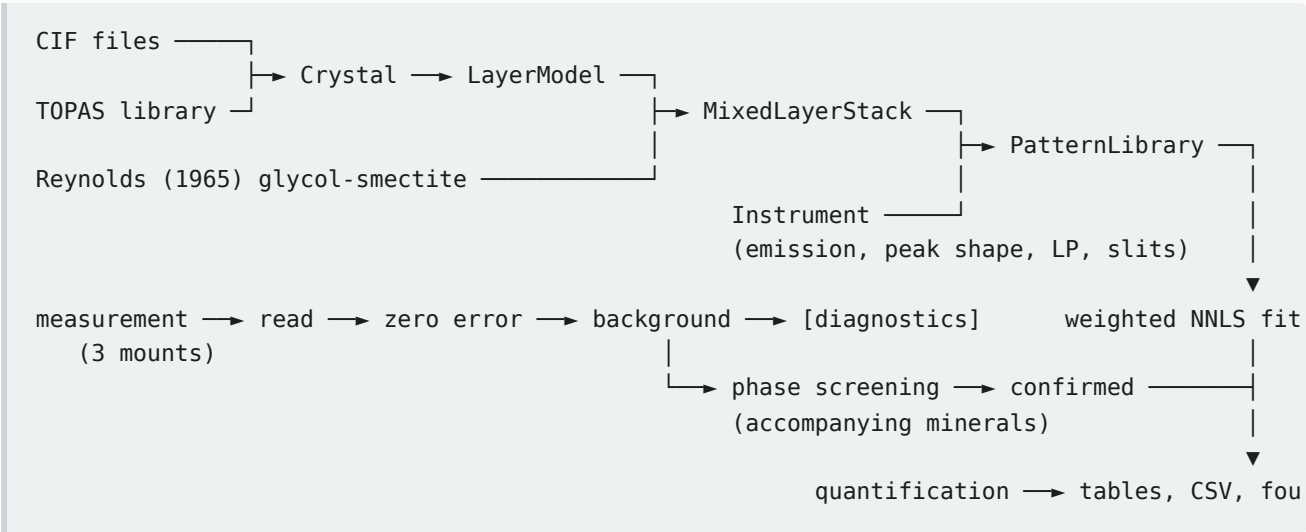
Results are given on two bases: over all fitted phases, and over the clay minerals alone, renormalised to 100 % with the accompanying minerals removed. The latter is the customary basis for reporting a clay mineral assemblage.

3 Implementation

3.1 Module map

Module	Contents
<code>crystal.py</code>	CIF reading, symmetry expansion, the <code>Crystal</code> and <code>LayerModel</code> classes, structure factors, derivation of a layer model from a polytype cell
<code>scattering.py</code>	Waasmaier–Kirfel scattering factors, species normalisation, displacement factors
<code>emission.py</code>	emission profiles, including the five-line Cu K α description
<code>optics.py</code>	Lorentz–polarization, March–Dollase, beam overflow
<code>profile.py</code>	pseudo-Voigt peak shapes, Caglioti widths, Scherrer broadening
<code>mixed_layer.py</code>	the matrix method, transition matrices, the CSDS and its pair weights
<code>pattern.py</code>	Instrument, reflection enumeration, powder, basal and mixed-layer pattern assembly, peak lists
<code>library.py</code>	the <code>PatternLibrary</code> , its builder, storage, and the spanning of orientation, thickness and spacing
<code>models.py</code>	the expected CIF sources, the structure search path, the Reynolds (1965) glycol–smectite layer
<code>bern.py</code>	import of a TOPAS/jEdit structure library into a phase database; clay/non-clay classification
<code>io.py</code>	readers for XRDML, <code>.xy</code> , Bruker <code>.raw</code> and two-column text
<code>calibration.py</code>	2 θ zero error on the quartz 100 reflection, merit profile for the slider
<code>background.py</code>	background components, peak stripping, the non-parametric background
<code>diagnostics.py</code>	kaolinite collapse and expandable-clay swelling
<code>detection.py</code>	finding accompanying minerals, by competitive fit or by peak matching
<code>nnls.py</code>	the weighted non-negative least-squares fit and its metrics
<code>quantification.py</code>	grouping by phase, the two normalisation bases, CSV export
<code>plots.py</code>	the four result figures and the report writer
<code>gui/</code>	the browser interface and its session state

3.2 Data flow



3.3 The reference library

Because the fit selects among pre-computed patterns rather than refining parameters, the library must span the parameters that actually vary between specimens. Three are spanned:

Dimension	Values	Why it must be spanned
Preferred orientation r	0.1 to 1.0 in steps of 0.1	texture is a property of the preparation, not of the mineral; it sets the ratio of basal to non-basal intensity
Mean crystallite thickness $\langle N \rangle$	5, 15, 50 layers	sets basal peak width; 10 layers give a 001 width near 0.8° where a well-crystallised illite shows 0.12°
Layer spacing d	illite 9.95, 10.02, 10.10 Å; chlorite 14.20, 14.35 Å	illite $d(001)$ varies with interlayer potassium and hydration; an error of 0.07 Å displaces the 001 by half a peak width

Together with the compositional series this gives, with default settings, 1601 patterns:

Phase	Patterns	Composition range
Illite/smectite (I/S)	1350	15 compositions, 0.20/0.80 to 0.99/0.01
Chlorite/smectite (C/S)	180	0.95/0.05, 0.90/0.10, 0.85/0.15
Illite	30	discrete
Chlorite	20	discrete
Kaolinite 1M, 2M	10 + 10	discrete
Glycol-smectite	1	pure end member

Each pattern is stored normalised to unit maximum *above* $4^\circ 2\theta$, not to its global maximum. Intensity rises steeply towards zero angle — the Lorentz factor still diverges after the beam-overflow correction — and no clay basal

reflection of interest lies below about 4° ; normalising on that rise would shrink the diffraction peaks themselves and leave the fit poorly conditioned.

Each pattern also has its own smooth continuum stripped before storage. An interstratified stack of finite thickness scatters a genuine diffuse continuum rising towards zero angle, but in a measurement that continuum is inseparable from air scatter and the direct beam and is always treated as background. Leaving it in the basis patterns would let a phase absorb the measured background and inflate its apparent share.

3.4 Numerical notes

- Symmetry operations are parsed by an explicit tokeniser, never by evaluating file content as code.
- The matrix sum (2.13) is truncated at the largest N carrying appreciable weight; the 2×2 matrix powers are accumulated once for all grid points simultaneously.
- Reflection enumeration is bounded by the measurable $d_{\min}$ at the longest wavelength in the emission profile.
- The NNLS solve is that of Lawson & Hanson as implemented in SciPy, applied to the row-weighted system; with 1600 columns and about 1800 points a fit takes well under a second.

4 Installation and structure data

4.1 Installation

Requirements

Component	Requirement	Notes
Python	3.10 or later	developed and tested on 3.11
NumPy	≥ 1.24	required
SciPy	≥ 1.10	required, for the NNLS solver
Dash	≥ 2.14	extra gui; needed only for the interface
Plotly	≥ 5.18	extra gui; also used by the figure export
gemmi	≥ 0.6	extra import; needed only to import a TOPAS structure library
pytest	≥ 7.4	extra dev; needed only to run the test suite
Operating system	any	pure Python; no compiler required. Linux, macOS and Windows
Disk	~ 200 MB	dependencies, plus 30–60 MB per stored reference library
Memory	~ 1 GB	a default library holds 1601 patterns on the measurement grid
Browser	any current	the interface runs locally at <code>127.0.0.1</code> ; nothing is sent over the network

Procedure

1. Obtain the source and create an isolated environment. A virtual environment is strongly recommended so that ClayQuant's dependencies do not disturb other Python installations.

```
git clone https://github.com/lomalaspina/ClayQuant.git
cd ClayQuant
python -m venv .venv
```

2. Activate the environment.

```
# Linux and macOS
source .venv/bin/activate

# Windows, PowerShell
.venv\Scripts\Activate.ps1

# Windows, cmd.exe
.venv\Scripts\activate.bat
```

3. Install. The extras select optional components; installing all of them is normal for interactive use.

```
pip install -e ".[gui,import,dev]"
```

The editable install (-e) means edits to the source take effect without reinstalling. To install only what is needed to compute patterns from a script, omit the extras: `pip install -e .`

4. Verify the installation. The test suite runs in a few seconds.

```
pytest -q
```

Tests that need structure data are skipped automatically when it is absent, and report as `s` rather than as failures; see §4.2. With no structure data installed, expect 29 passed and 10 skipped; with the four CIFs in place, 45 passed. Any *failure* indicates a problem and should be reported.

5. Provide structure data (§4.2), build a reference library (§4.4), and start the interface:

```
clayquant-gui
```

which prints a local address, by default `http://127.0.0.1:8050`, to open in a browser. Use `--port` to choose another port and `--host` to bind an interface other than the loopback.

Upgrading. After pulling a new revision, re-run `pip install -e ".[gui,import,dev]"` to pick up any dependency change, and rebuild the reference library if the pattern calculation has changed. Stored libraries carry the metadata of the settings they were built with, and a library built by an older revision is read but may not correspond to the current calculation.

4.2 Structure data and licensing

No structure data is distributed with ClayQuant. The crystal structures it uses are ICSD entries, which are licensed by FIZ Karlsruhe and may not be redistributed. They must be exported from your own subscription. The repository's ignore rules are written so that a local copy cannot be committed by accident.

Export each of the following as CIF and save it under exactly the name given:

Phase	ICSD	File name	Structure reference
Illite	90144	illite_ICSD_90144.cif	Gualtieri (2000)
Chlorite (clinochlore IIb-4)	164234	chlorite_ICSD_164234.cif	Zanazzi, Comodi, Nazzareni & Andreozzi (2009)
Kaolinite 1M	63192	kaolinite_1M_ICSD_63192.cif	Bish & Von Dreele (1989)
Kaolinite 2M	30285	kaolinite_2M_ICSD_30285.cif	Gruner (1932)

The directory is searched in this order:

1. the path in the environment variable `CLAYQUANT_STRUCTURE_DIR`, if set;
2. a `structures` folder in the current working directory;
3. the package data directory.

To check what has been found:

```
python -c "from clayquant.models import available_phases, structure_directory; \
print(structure_directory()); print(available_phases())"
```

The glycol–smectite layer requires no file. It is the published one-dimensional model of Reynolds (1965), transcribed into the source with its provenance, and is reproduced under §A.2.

4.3 Importing a TOPAS structure library

The accompanying non-clay minerals — quartz, feldspars, carbonates, sulphates, oxides — come from a structure library of your own. ClayQuant reads the jEdit macro-menu form in which TOPAS structure collections are commonly distributed: an XML file whose items each insert a TOPAS `str` block with a phase name, cell parameters, space group and site lines.

```
clayquant-import-structures my_structures.xml -o structures/phases.json
```

Space group symbols are expanded to explicit symmetry operations with `gemmi`, including the origin-choice settings that TOPAS writes as a trailing letter (`Fd-3mS` for origin choice 1, `Fd-3mZ` for choice 2; see §A.3). The operations are stored in the generated database, so `gemmi` is not needed afterwards.

The generated database is derived from licensed structure data and is likewise kept out of version control.

4.4 Building the reference library

A library is built once for a given instrument configuration and measurement grid, and reused. Build it on the same 2θ grid as the measurements.

```
clayquant-build-library -o library/clays.npz
```

Building the default library takes about three minutes. Within the interface the same operation is available as *Build library now*, using the current instrument settings.

5 Operating procedure

5.1 Specimen preparation assumed

ClayQuant assumes the preparation customary for clay mineral analysis: the bulk sample is ground, the clay fraction separated by sedimentation or centrifugation, and the separate deposited on glass slides to give oriented mounts, of which one is left air-dried, one solvated with ethylene glycol and one heated, typically to 500 °C. A bulk (randomly oriented) measurement of the whole sample is useful context but is not required by the program.

5.2 Step 1 — Load

Enter the folder holding the measurements and press *Scan folder*. Files are listed, and the three mounts are assigned from the drop-down menus; names containing `air`, `eg` or `glyc`, and `heat` are pre-selected where recognised. Press *Load*.

The point count, angular range and step are reported for each mount. Check that all three cover the same range: the diagnostics compare mounts point by point after interpolation, and a mount measured over a different range will still work but will restrict the comparison.

5.3 Step 2 — Zero error

The 2θ zero error is determined on the quartz 100 reflection, $d = 4.2551 \text{ \AA}$, $2\theta = 20.859^\circ$ for $\text{Cu K}\alpha_1$. This reflection is chosen because no clay basal reflection falls on it.

Find peak automatically locates it and sets the slider; the slider itself steps in 0.01° , half the usual measurement step, and the merit curve beneath shows the figure of merit against candidate offsets so the chosen value can be seen to be the optimum. Each mount carries its own zero error, because each is a separately prepared slide with its own displacement; set all three.

Quartz 100 is only 0.4° from the kaolinite 020 band at $4.36 \text{ \AA}$. In a clay separate quartz is often weak while kaolinite is strong, and a wide search window will find kaolinite and return a zero error of about

-0.4° , displacing the whole pattern. The search window is therefore deliberately narrow ($\pm 0.30^\circ$ by default) and a signal-to-noise test is applied. Where quartz is genuinely absent the program reports that it found no calibration peak and leaves the shift at zero rather than returning a meaningless number; set the value by hand or calibrate on another reflection. See §A.5.

5.4 Step 3 — Background

Choose a model and adjust it until it follows the background without cutting into the peaks. The components are described in §6; the $1/x$ term, which describes the low-angle rise from air scatter and the direct beam, can be added to any of the others. A non-parametric peak-stripped background is also available for backgrounds no low-order function follows.

Models are fitted not to the raw counts but to a peak-stripped estimate of the background, then refined asymmetrically. This matters: a clay pattern has no peak-free points below about 8° , so fitting the raw intensities there drags the background up into the basal reflections and removes the very signal being quantified. Peak stripping errs the other way, underestimating the background beneath a broad strong low-angle reflection, which is the safe direction but means the low-angle background should always be checked by eye.

Apply to all mounts uses the same model for the three mounts, which is appropriate when they were measured under identical conditions.

5.5 Step 4 — Main minerals

Load the phase database of §4.3 and press *Search for main minerals*. If a reference library has been loaded, every candidate is calculated and fitted in competition with the clay patterns and with each other in a single non-negative least-squares solve, and each is ranked by the share of the pattern it takes. Without a library, the program falls back to matching predicted peak positions, which is a much weaker test (§A.6).

A dialog lists what was found:

Evidence for the following main minerals has been found in the data:

with, for each, its share of the pattern, how many of its expected lines were found, and the best signal-to-noise among them. Phases with a clear share are ticked; the tail at one or two per cent is listed but left unticked, being largely least-squares slack rather than mineralogy.

This is a shortlist for judgement, not an identification. Phases with similar cells — the feldspars, the spinels, dolomite against ankerite — cannot be separated on diffraction evidence alone at these levels, and a database assembled for general laboratory use contains many phases that cannot occur in a sediment. Tick what the sample can plausibly contain. Selected phases are fitted with the clays and reported separately, then removed when the clay percentages are normalised.

5.6 Step 5 — Kaolinite

Kaolinite is identified by the loss of its 7.15 Å reflection on heating. Because the two mounts are separate slides with different amounts of material, the comparison is meaningless until they are put on a common scale; the program does this on a reflection that survives heating, by default the illite 10 Å reflection, and reports the scale factor used.

The result is the percentage of the 7.15 Å area lost. What survives is chlorite 002, which occupies the same position and does not collapse. A sample with both minerals therefore shows partial collapse, and the surviving fraction is an estimate of the chlorite contribution to that peak.

5.7 Step 6 — Expandable clay

Expandable layers are identified by the shift of the low-angle reflection on glycol solvation: a smectite interlayer takes up a two-layer glycol complex and expands to about 16.9 Å. The program locates the reflection in a window of the air-dried and glycolated mounts, reports both positions in ångström and the shift between them, and the ratio of the areas.

A shift towards larger d is the diagnostic. A shift of 13.8 → 16.1 Å, for example, is unambiguous evidence of expandable layers. No shift means either that there are none, or that the expandable component is a minor interstratified fraction whose effect is spread through the illite/smectite series rather than concentrated in a discrete peak — which the quantitative fit will still detect.

5.8 Step 7 — Fit and results

Load a reference library, choose the mount to fit — normally the glycol-solvated one — set the angular range, and press *Run NNLS fit*.

Choose the fitting range deliberately. The lower limit should exclude the direct-beam tail, and 4° is a reasonable default; setting it lower rarely adds information and can let the fit chase background. The upper limit should include the region where the main minerals have their strong lines, so that they are properly determined, but there is little clay information above about 35°.

Restrict orientation parameters limits the fit to library patterns of particular March–Dollase values. Since a well-prepared oriented mount is strongly textured, restricting to the low values (0.1–0.3) both speeds the fit and prevents it from using near-random patterns to model what is really a texture effect.

The results appear as four plots and two tables, described in §8. *Export tables and plots* writes them to disk.

Read the plots, not only the table. The fit will always return numbers. Whether they mean anything is visible in the difference curve of the *Fit* plot and in whether the per-phase curves of the *All phases* plot sit under real features of the measurement. A large systematic residual at one strong peak usually means a layer spacing or crystallite thickness outside the range the library spans (§9).

5.9 The simulator

The simulator drives the calculation directly, independently of any measurement, for teaching, for exploring what a given composition looks like, and for checking the effect of a parameter. It computes a single interstratified or discrete pattern from the controls of §6 and displays it immediately.

6 Reference: interface controls

Every control is listed below by the tab on which it appears, with the identifier used internally, its default, and its meaning. Sliders show their current value; range sliders have two handles.

6.1 Tab 1 — Load

Control	Identifier	Default	Meaning
Data folder	directory	—	Folder searched for measurement files. Sub-folders are not searched.
Scan folder	scan	—	Lists readable files and offers them in the three drop-downs. Names containing <i>air</i> , <i>eg/glyc</i> and <i>heat</i> are pre-selected where recognised.
Air-dried / Ethylene glycol / Heated	file-air, file-glycol, file-heated	—	Assignment of a file to each mount. The glycol mount is the one normally fitted; the other two drive the diagnostics.
Load	load	—	Reads the selected files and reports points, range and step for each.

6.2 Tab 2 — Zero error

Control	Identifier	Default	Meaning
Mount	zero-mount	Air-dried	Which mount the slider and the automatic search act on. Each mount has its own zero error.
Calibration reflection	zero-reference	Quartz 100, 4.2551 Å	The reflection whose known position defines the zero. Quartz 100 clears all clay basal reflections. Quartz 101 (3.3435 Å) is stronger but overlaps illite 003.
Zero error (°2θ, steps of 0.01)	zero-slider	0.00	The offset <i>subtracted</i> from the measured angles. The step is 0.01°, half the usual 0.02° measurement step.
Find peak automatically	zero-auto	—	Locates the calibration peak within a narrow window and sets the slider. Reports the signal-to-noise, or states that no calibration peak was found.

6.3 Tab 3 — Background

Control	Identifier	Default	Meaning
Mount	bg-mount	Air-dried	Mount displayed and fitted.
Main component	bg-kind	Chebyshev	<i>Polynomial</i> : $\Sigma c_k u^k$ in the angle scaled to $[-1,1]$. <i>Chebyshev</i> : $\Sigma c_k T_k(u)$, better conditioned at high degree and the usual Rietveld choice. <i>Exponential</i> : a single decaying term. <i>Peak-stripped</i> : non-parametric; the stripped estimate is used directly and follows any shape, at the cost of having no parameters.
Degree	bg-degree	5	Degree of the polynomial or Chebyshev component; degree n contributes $n+1$ terms. Raise it only as far as the background demands: an over-flexible background follows peaks.
Exponential decay (1°)	bg-decay	0.15	Decay constant of the exponential component, in reciprocal degrees.
Add $1/x$ term	bg-inverse	on	Accumulates $c/(2\theta+x_0)$ on top of the main component. This describes the low-angle rise from air scatter and the direct beam and can be combined with any of the others.
$1/x$ offset ($^\circ 2\theta$)	bg-offset	1.0	The offset x_0 , which keeps the term finite as the angle approaches zero.
Peak-stripping width ($^\circ 2\theta$)	bg-snip	4.0	Width of the widest feature the stripping treats as a peak. Too small and broad reflections are treated as background; too large and the background itself is followed poorly.
Apply to this mount / Apply to all mounts	bg-apply, bg-apply-all	—	Stores the fitted background. <i>All</i> uses the same model for the three mounts, appropriate when they were measured under identical conditions.

6.4 Tab 4 — Main minerals

Control	Identifier	Default	Meaning
Phase database (.json)	db-path	structures/phases.json	The database produced by <code>clayquant-import-structures</code> .
Load database	db-load	—	Reads it and reports how many phases it holds. Phyllosilicates in the database are left to the clay library and are not offered here.
Mount to search	detect-mount	Air-dried	Which mount is searched. The air-dried mount is usually the best choice: glycol dilutes the accompanying minerals and heating alters some of them.
Unit cell allowance (%)	detect-allowance	2	Relative cell deviation allowed when matching positions, applied as one <i>coherent</i> scaling of all of a phase's d-spacings. Most accompanying minerals are solid solutions whose cells differ from the database entry. Applied per peak instead of coherently, a 2 % allowance is a window up to 1.6° wide and matches almost anything (§A.6).
Minimum share (‰) or score (%)	detect-score	10	Reporting threshold. With a library loaded it is read as parts per thousand of the pattern (10 = 1 %); without one, as a percentage score of the peak-matching test.
Minimum peak S/N	detect-snr	5	Height above the local noise a predicted line must reach to count as found. Used by the peak-matching fallback and by the per-line evidence shown in the dialog.
Search range (°2θ)	detect-range	4 to 40	Angular range considered. It should cover the strong lines of the expected accompanying minerals.
Search for main minerals	detect-run	—	Runs the search and opens the confirmation dialog.
Include selected in the fit / Dismiss	detect-accept, detect-dismiss	—	Confirms the tick list, or discards it. Nothing is added to the fit until confirmed.

6.5 Tab 5 — Kaolinite

Control	Identifier	Default	Meaning
7.15 Å window (°2θ)	kao-window	11.6 to 13.2	Integration window for the kaolinite 001 / chlorite 002 reflection.
Scaling reference	kao-reference	Illite 10 Å	The reflection used to put the air-dried and heated mounts on a common scale. It must survive heating. <i>Manual</i> uses the factor below instead.
Manual scale factor	kao-scale	1.00	Used when the scaling reference is set to manual.
Compare air and heated	kao-run	—	Reports the percentage of the 7.15 Å area lost, the surviving fraction attributable to chlorite 002, both areas, and the scale factor used.

6.6 Tab 6 — Expandable

Control	Identifier	Default	Meaning
Search window (°2θ)	sme-window	4.0 to 10.0	Window searched for the low-angle reflection in both mounts. It must be wide enough to hold the air-dried position and the glycolated position, which may differ by several degrees.
Scaling reference	sme-reference	Illite 10 Å	As for kaolinite: puts the two mounts on a common scale so that the area ratio is meaningful.
Manual scale factor	sme-scale	1.00	Used when scaling is manual.
Compare air and glycol	sme-run	—	Reports both positions in ångström, the shift, and the area ratio.

6.7 Tab 7 — Fit and results

Control	Identifier	Default	Meaning
Library file (.npz)	lib-path	library/clays.npz	The stored reference library.
Load library / Build library now	lib-load, lib-build	—	Loads a stored library, or builds one with the current instrument settings. Building takes a few minutes.
Mount to fit	fit-mount	Ethylene glycol	Which mount is quantified. The glycolated mount is the standard choice because glycolation resolves the illite/smectite series.
Fit range (°2θ)	fit-range	4 to 34	Range fitted. Exclude the direct-beam tail at the low end; there is little clay information above about 35°.
Restrict orientation parameters	fit-orientations	all	Limits the fit to library patterns of the chosen March–Dollase values. Restricting to low values is appropriate for a well-oriented mount and prevents the fit using near-random patterns to model a texture effect.
Subtract the fitted background	fit-subtract	on	Subtracts the background stored in step 3. Turn it off only if the measurement is already background-corrected.
Run NNLS fit	fit-run	—	Performs the weighted non-negative least-squares fit and produces the plots and tables.
Export folder / Export tables and plots	export-path, export-run	results	Writes the outputs of §8.

6.8 Simulator

Control	Identifier	Default	Meaning
Model	sim-model	Interstratified	<i>Interstratified</i> computes a two-component stack; <i>Discrete phase</i> computes the pure phase from its structure.
Phase / host	sim-phase	illite	The discrete phase, or the host layer type A of the interstratification (the other component being glycol–smectite).
Host layer fraction	sim-fraction	0.80	W_A , the proportion of host layers. 1.0 is the pure host; 0.0 the pure smectite.
Junction probability P(host → smectite)	sim-pab	0 (random)	P_{AB} of §2.7. Zero selects random stacking ($R = 0$); values above W_B give ordering, below it segregation.
Mean crystallite thickness (layers)	sim-csds	10	$\langle N \rangle$ of the lognormal CSDS. This is the principal control on basal peak width.
CSDS width β (in $\ln N$)	sim-beta	0.35	Width of the distribution in $\ln N$. Larger values give a broader spread of thicknesses and smoother peak profiles.
March–Dollase orientation r	sim-po	0.3	Texture parameter. Recall that it does not change the shape of a purely basal series, only the basal-to-non-basal ratio.
Glycolated smectite $d(001)$ (Å)	sim-dsmectite	16.86	Repeat distance of the glycol–smectite layer. 16.86 Å is the value measured by Reynolds (1965); 16.9–17.1 Å is also used.
2θ range (degrees)	sim-range	2 to 40	Calculated range.
Options	sim-options	—	Toggles for including the non-basal contribution and for showing the component curves separately.

7 Reference: command line

7.1 clayquant-build-library

Option	Default	Meaning
-o, --out	library/clays.npz	Output file.
--two-theta-min, --two-theta-max	2, 40	Angular range of the calculated grid, in degrees.
--step	0.02	Step of the calculated grid. Match the measurement.
--csds-means	5 15 50	Mean crystallite thicknesses, in layers; one set of patterns per value.
--csds-beta	0.35	Width of each lognormal distribution in $\ln N$.
--orientations	0.1 ... 1.0	March–Dollase parameters to span.
--quiet	off	Suppress progress reporting.

7.2 clayquant-import-structures

Option	Default	Meaning
xml	—	The jEdit macro-menu XML file to import (positional).
-o, --out	structures/phases.json	Output phase database.
--quiet	off	Suppress the per-phase notes and the failure list.

7.3 clayquant-gui

Option	Default	Meaning
--port	8050	Port to serve on.
--host	127.0.0.1	Interface to bind. The default is the loopback, so the interface is not reachable from the network.
--debug	off	Enable the development server's reloader and error pages.

8 Output files

Export tables and plots writes seven files into the chosen folder.

File	Contents
clayquant_mounts.html	The three mounts overlaid: air-dried, glycol-solvated, heated. The plot in which the kaolinite collapse and the glycol expansion are seen directly.
clayquant_fit.html	The measurement, the fitted curve and the difference, with R_{wp} in the title.
clayquant_phases.html	The measurement with every fitted phase drawn separately, and their sum.
clayquant_clay_phases.html	The same for the clay minerals alone, annotated with the clay share of the whole pattern.
clayquant_all_phases.csv	Quantification over all fitted phases.
clayquant_clays_only.csv	Quantification over the clay minerals alone, renormalised to 100 %.
clayquant_curves.csv	The numerical curves: 2 θ , measured, fit, and one column per phase, so the figures can be redrawn in any other program.

The figures are self-contained HTML: they need no plotting software to open and keep the zoom and hover of the interactive versions.

8.1 Columns of the quantification tables

Column	Meaning
phase	Phase name. Interstratified series appear as I/S and C/S, summed over the individual library patterns that contributed.
group	clay or main. Determines which phases the clay-basis normalisation keeps.
coefficient	Sum of the fitted scale factors c_p of that phase's patterns. Depends on the arbitrary normalisation of the library and is meaningful only in comparison within one fit.
scattering_percent	The phase's share of the calculated integrated intensity. The primary number.
amplitude_percent	The phase's share of the summed scale factors.
entries	The individual library patterns that took non-zero coefficients, naming the composition, orientation parameter, crystallite thickness and layer spacing selected. This is where the fit reports what it found: an I/S share carried by 0.90/0.10 patterns at $d = 9.95 \text{ \AA}$ and $N = 50$ is a specific mineralogical statement.

Each file carries a header recording the measurement, R_{wp} and R_p , the basis of normalisation, the clay share of the whole pattern where applicable, and a statement that the percentages are shares of diffracted intensity and not weight percent, so that a file cannot later be mistaken for a mass analysis.

9 Limitations

These are properties of the method, not defects to be worked around. Each is stated with its practical consequence.

1. **A fixed library cannot refine.** Peak positions, widths and orientation are sampled at the values the library was built with. A specimen whose layer spacing or crystallite thickness falls outside the spanned range will fit badly,

and the misfit appears as a systematic residual at a strong peak rather than as an obviously wrong answer. Widen the spanned ranges and rebuild if this is seen.

2. **Percentages are shares of intensity, not of mass** (§2.13).
3. **Kaolinite polytypes are not distinguishable from basal reflections** (§2.3). A division of intensity between $1M$ and $2M$ is not a polytype determination.
4. **The non-basal contribution of an interstratified phase is approximated** by that of its host (§2.12). Turbostratic hk bands are not modelled with their true sawtooth profile. The approximation is good for a strongly oriented mount and degrades as $r \rightarrow 1$.
5. **Only two-component interstratifications are treated.** Three-component systems, and mixed-layer chlorite/smectite/illite, are outside the model.
6. **Accompanying minerals of similar cell cannot be separated.** The screen ranks candidates by explained intensity, which is far sharper than position matching, but the feldspars, the spinels and the rhombohedral carbonates remain close enough that the ranking should be read as a shortlist for judgement (§5.5).
7. **The zero error requires quartz.** Where quartz is absent the program declines to report a value; it must then be set by hand, and an error there displaces the whole pattern and corrupts the fit.
8. **The low-angle background is the weakest part of the reduction.** There are no peak-free points below about 8° , so the background there is an extrapolation. It should always be inspected by eye, particularly when a smectite 001 near 5.2° is at issue.
9. **The ordered case is implemented but not spanned by the default library** (§2.7). Regularly interstratified minerals such as rectorite and corrensite will not be fitted correctly by a library built with the default settings.
10. **No absorption or microabsorption correction is applied.** For an infinitely thick specimen in Bragg–Brentano geometry the absorption factor is independent of angle and cancels in relative intensities; a thin mount violates this, as does a specimen of strongly contrasting absorption between phases.

10 References

The list below covers the physical model, the numerical methods, the software libraries, and the four crystal structures on which ClayQuant's clay mineral calculations rest. The accompanying mineral structures come from a user-supplied library and are not covered here, because their citations belong to the database they were drawn from; every one of them is identified by database and entry number in Appendix D, from which the citation can be retrieved.

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Appendix A — Record of verification tests

This appendix records the tests by which ClayQuant was verified during development: what each test was for, how it was done, what it returned, and what was concluded. Several tests are recorded that *failed* and led to a change in the program; these are the more informative ones and are reported in full.

On the measurements used. Development was verified against 33 measurements of eleven clay separates from a working laboratory. Those measurements are proprietary and are **not** reproduced here, nor distributed with the program. They are referred to below as Specimen 1, Specimen 2 and so on, and only derived quantities — agreement factors, peak widths, fitted positions — are quoted, never the measured data themselves.

A.1 The interstratification model against closed-form results

Reasoning. The matrix method of §2.5 is the core of the program and is easy to implement subtly wrongly: the phase convention of equation (2.10) can be transposed, the pair counting of equation (2.11) can be off by one, and the stationary distribution can be inconsistent with the transition matrix. None of these produce obviously absurd output — they produce a plausible pattern that is wrong. The model must therefore be tested against cases whose answer is known independently.

Method and results. Twenty-five tests (`test_mixed_layer.py`):

Test	Reasoning	Result
Single component reproduces the Laue interference function, for $N = 1, 2, 5, 17$	With one layer type the general result must collapse to the closed form (2.7). This checks the phase convention, the pair counting and the N -dependence simultaneously, against an exact expression.	Agreement to numerical precision at all four thicknesses. Pass.
Two identical layer types behave as one	Setting A and B to the same layer must give the one-component answer whatever the proportions, which tests that the two components are combined without double counting.	Identical to the single-component calculation. Pass.
A pure end member ignores the absent layer	At $W_A = 1$ the pattern must not depend on what B is. A failure here would show the absent component leaking through the transition matrix.	Independent of the B layer model. Pass.
Single-layer crystallites show no interference	At $N = 1$ the intensity must be the layer structure factor squared, with no interference term at all. This isolates the pair sum.	Reduces to $\sum W_m F_m ^2$. Pass.
Intensity is non-negative — 15 cases: random, ordered and segregated stacking at $W_A = 0.05, 0.2, 0.5, 0.8, 0.95$	Intensity is a squared modulus and cannot be negative anywhere. A sign error in the real part of the matrix sum, or an inconsistent stationary distribution, shows up as small negative excursions between peaks. This is the most sensitive cheap test of the whole formalism.	Non-negative throughout in all 15 cases. Pass.
CSDS pair weights	$G(n)$ of equation (2.14) must equal 1 at $n = 0$ and decrease monotonically to zero, or the thickness distribution is misapplied.	As specified. Pass.
The random transition matrix is the Markov special case	Building a matrix with $P_{AB} = W_B$ through the general Markov constructor must give exactly the random one. This checks the constructor against its own special case.	Identical matrices. Pass.
Impossible ordering is rejected	A P_{AB} implying a probability outside $[0,1]$ is unphysical and must raise rather than silently produce nonsense.	Raises. Pass.

Conclusion. The interstratification model reproduces every case for which an independent answer exists, and satisfies the positivity constraint across the full range of composition and ordering.

Consistency of the basal and three-dimensional calculations

Reasoning. The same phase can be computed two ways — as a three-dimensional structure through equation (2.3), or as a one-dimensional layer stack through §2.5. These are independent code paths that must agree on the basal series of a pure phase, otherwise the mixed-layer patterns and the discrete patterns are on different scales and the fit apports intensity wrongly between them.

Results (`test_basal_consistency.py`): for illite, chlorite, kaolinite 1*M* and kaolinite 2*M*, the basal series computed from the one-dimensional layer model matches the 00*l* intensities of the full three-dimensional calculation; the derived layer thickness equals $d(001)$ of the cell divided by the number of layers per cell; and the absolute spacings agree with the accepted values (10.02, 14.28, 7.15, 7.15 Å). **Pass.**

Conclusion. The polytype folding of §2.3 is exact for the structures used, and the two code paths are on a common scale.

A.2 Validation against Reynolds (1965)

Reasoning. Everything above is internal consistency: the program agreeing with itself. An external check against a published experiment is needed, and the glycol–smectite complex is the ideal one, because Reynolds (1965) published both his structural model (his Table 1) and his observed and calculated structure factors and intensities for fourteen basal orders (his Table 2). Reproducing his calculated quantities from his stated model tests the layer structure factor of equation (2.6), the scattering factors, the displacement factors and the Lorentz–polarization treatment together, against data that did not come from this program.

Method. The model of his Table 1 was transcribed, the basal structure factors computed from equation (2.6), converted to intensities with the random-powder Lorentz–polarization factor he states he used, normalised at 001, and compared with his Table 2. Agreement is measured by his own reliability factor, $R = \Sigma ||F_o| - |F_c|| / \Sigma |F_o|$.

An ambiguity, and how it was resolved. His Table 1 gives occupancies per atomic plane, but the octahedral plane lies on the mirror at $z = 0$ and is quoted as half its content; and his glycol rows are labelled as half-molecules (" H_2C-HO "), which may or may not include their hydrogens. Four readings are possible. Rather than guess, all four were computed and compared with his published values:

Reading	Electrons per cell	$R(F_o)$	Mean $ \Delta \log_{10} I $
A — octahedral plane as quoted, glycol without H	452	0.299	0.350
B — octahedral plane as quoted, glycol with H	476	0.312	0.327
C — octahedral plane doubled, glycol without H	480	0.077	0.158
D — octahedral plane doubled, glycol with H	504	0.039	0.028

Result. Reading D is correct by a wide margin. It also reproduces independently the electron count Reynolds states for his cell ("about 500 electrons"): 504 against his ~ 500 , where the other readings give 452 to 480. Its reliability factor, $R = 0.039$, is slightly *better* than the $R = 0.042$ he reports for his own best model — as expected, since modern scattering factors are used. Four tests now assert this permanently (`test_reynolds_1965.py`): the electron count, the reliability factor, the fourteen calculated intensities, and the ordering of the strong orders.

Conclusion. The one-dimensional structure factor calculation reproduces a published experimental analysis to better than its original agreement. This is the strongest external validation in the program, and it simultaneously confirms the choice of Lorentz–polarization factor of §2.8, since no other choice reproduces these intensities.

A.3 Import of a TOPAS structure library

Reasoning. The accompanying minerals enter the fit as calculated patterns, so an error in reading their structures — a misparsed cell parameter, a wrong space group setting, a mis-assigned occupancy — produces a wrong pattern that the fit will nevertheless use. The import must be verified against known diffraction data, phase by phase.

Method and results. A library of 205 phases was imported and the strongest calculated line of each of sixteen well-known phases compared with its accepted position.

Phase	Calculated (°2 θ)	Accepted (°2 θ)	Δ
Quartz	26.64	26.64	+0.00
Calcite	29.40	29.40	+0.00
Rutile	27.44	27.44	+0.00
Magnetite	35.43	35.44	−0.01
Halite	31.69	31.70	−0.01
Hematite	33.16	33.15	+0.01
Microcline	27.51	27.50	+0.01
Dolomite	30.94	30.96	−0.02
Albite	27.92	27.94	−0.02
Anhydrite	25.42	25.44	−0.02
Anatase	25.31	25.28	+0.03
Siderite	32.02	32.05	−0.03

Four phases appeared at first to disagree and were examined individually:

- **Barite** and **fluorite**: the calculation was right and the comparison value wrong. Barite's strongest line is 3.10 Å (28.74°), not the 3.44 Å often quoted from memory, and the full calculated list — 3.445, 3.318, 3.104, 2.835, 2.122, 2.107 Å with intensities 98, 63, 100, 53, 81, 56 — matches the powder file essentially exactly. Fluorite's 111 and 220 differ by only a few per cent in intensity and their order is not diagnostic.
- **Gypsum**: the calculation gives 4.28 Å as strongest where powder files give 7.63 Å. This is the well-known preferred-orientation artefact of gypsum powder files, caused by its perfect (010) cleavage; the source library carries an explicit preferred-orientation direction for gypsum for exactly this reason. The calculated *random-powder* pattern is correct as computed.
- **Pyrite**: positions match exactly (3.128, 2.709, 2.423, 2.212, 1.916, 1.634 Å); only the relative intensities of 220 and 311 were in question, and published values differ between sources.

Determining the TOPAS origin-choice convention

Reasoning. Thirteen phases failed to import because TOPAS writes the two origin choices of centrosymmetric cubic and tetragonal groups with a trailing letter — Fd-3mS against Fd-3mZ — rather than the 1/2 of the International Tables. Guessing the mapping was unacceptable: the two origins differ by a translation of $(\frac{1}{8}, \frac{1}{8}, \frac{1}{8})$, so the wrong choice silently produces wrong intensities for spinels, magnetite, zircon and anatase.

Method and result. The convention was determined from the structures themselves rather than assumed. Diamond, silicon and franklinite, written S, place their 8a site at (0,0,0), which is origin choice 1. Magnetite, spinel, chromite, hercynite and magnesioferrite, written Z, place it at $(\frac{1}{8}, \frac{1}{8}, \frac{1}{8})$, which is origin choice 2. Zircon (I4₁/amdZ) has Zr at $(0, \frac{3}{4}, \frac{1}{8})$, the standard choice-2 setting; anatase (I4₁/amdS) has Ti at the origin, choice 1. Eight independent structures agree: **S = origin choice 1, Z = origin choice 2**. With this mapping all 205 phases import.

Conclusion. Structure import is verified: positions reproduce accepted values to 0.03° across twelve chemically diverse phases, the four apparent discrepancies are explained and none is an error in the program, and the origin-choice convention is established from evidence rather than assumption.

A.4 Workflow tests on synthetic data

Reasoning. The reduction steps — zero error, background, diagnostics, fit — can only be tested for *correctness* against data whose answer is known in advance, which no real measurement provides. Synthetic patterns were therefore constructed from the program's own library with a known composition, a known zero error and a known background, and the workflow required to recover them.

Test	Reasoning	Result
Zero error is recovered	A known offset is imposed on a synthetic pattern; the calibration must return it. Tests the peak location and the sign convention — a sign error here would shift every pattern the wrong way.	Recovered within the 0.01° step. Pass.
Background with an accumulated $1/x$ term	The $1/x$ term must be combinable with any main component, as specified. Tests the design matrix assembly.	Recovers an imposed background of that form. Pass.
All background components can be combined	Polynomial, Chebyshev, exponential and $1/x$ simultaneously — the worst case for conditioning of the linear solve.	Fits without degeneracy. Pass.
NNLS recovers a known composition	The central claim of the method: a synthetic mixture of known proportions must be recovered. Tests the design matrix, the solver and the share computation.	Recovers the composition. Pass.
Kaolinite collapse is detected	A synthetic pair with kaolinite removed from the heated mount must be diagnosed as collapse.	Detected. Pass.
Chlorite survives heating	The complement: chlorite must <i>not</i> be reported as collapsed, or every chloritic sample would be called kaolinitic.	Survives. Pass.
Smectite swelling is detected	A synthetic pair with the expandable component shifted must be diagnosed as swelling.	Detected. Pass.

A.5 File readers and zero-error behaviour on real mounts

Readers

Reasoning. A reader that silently misreads — an offset by one point, a wrong step, an intensity scaling — corrupts everything downstream while appearing to work. The strongest available check is that two independent encodings of the same measurement, the instrument's native XRDML and its own exported two-column text, must agree exactly.

Result. All 33 XRDML files read, and for every specimen where both forms exist the two agree **bit for bit**: maximum difference 0.000 counts, identical 2 θ grids. The instrument configuration carried in the XRDML — wavelengths, goniometer radius, divergence slit, counting time — is also recovered, so a calculation can be set up to

match the measurement. An initial version failed on all 33 files because intensities are held in a <counts> element rather than <intensities>; both are now accepted, as are measurements holding several appended scans.

Zero error

Reasoning. The zero error was first determined with a wide search window ($\pm 0.6^\circ$). Run across all eleven specimens, this produced shifts ranging to -0.40° , with the three mounts of one specimen disagreeing by 0.33° . Mounts of the same specimen are separate slides, so they may differ by a few hundredths of a degree — but not by a third of a degree. The disagreement itself was the evidence that something was wrong.

Diagnosis. Inspecting the region directly showed that in the kaolinite-rich specimens there is no quartz 100 peak at all, and the routine was locking onto the kaolinite 020 band at $4.36 \text{ \AA}$, 0.4° away. Quartz 100 clears every clay *basal* reflection, which is why it is the standard choice, but it does not clear the kaolinite *hk* bands.

Result after correction (narrow window, signal-to-noise test):

Specimen	Air	Glycol	Heated	Spread
1	-0.06	-0.06	-0.04	0.02
2	-0.06	-0.06	-0.06	0.00
3	-0.04	-0.05	-0.05	0.01
4	-0.04	-0.06	-0.06	0.02
5	-0.04	-0.04	-0.04	0.00
6	-0.05	-0.03	not found	0.02
7	-0.02	+0.04	not found	0.06
8	-0.06	+0.00	-0.06	0.06
9	-0.05	+0.06	not found	0.11
10, 11	no calibration peak in any mount			—

Conclusion. Detected shifts now lie between -0.06° and $+0.06^\circ$, are consistent between the mounts of a specimen to 0.02° in the majority of cases, and are of the magnitude expected for instrument zero and specimen displacement. The eight mounts in which quartz is genuinely absent now report that fact instead of a fabricated number. *This test is the reason the search window is narrow and the signal-to-noise test exists*; with the original settings the program would have silently displaced whole patterns by 0.4° .

A.6 Development of the accompanying-mineral search

Reasoning. Searching a 205-phase database for the few minerals actually present is a discrimination problem, and a method that reports everything is worthless. Each candidate method was tested on a synthetic pattern of *known* content — quartz, calcite and albite in known proportions — so that false positives could be counted, and then on a real specimen.

Method	Reasoning	Result	Verdict
1. Per-peak window, 2 % cell allowance	The obvious first approach: allow each expected line to be found anywhere within the tolerance implied by a 2 % cell deviation.	190 of 205 phases scored 100 % on a three-phase pattern. A 2 % allowance is a window up to 1.6° wide at high angle; on a peak-rich pattern nearly every phase finds something near every expected line.	Rejected.
2. One coherent cell scaling + cosine similarity	A real cell deviation stretches the whole lattice together, so all of a phase's lines must move by the same relative amount — a far stronger constraint than independent windows. Intensity agreement was added as a cosine similarity.	Better: the three true phases ranked 2nd, 4th and 7th. But 157 phases still passed, and graphite, aluminium, copper and lead ranked at the top. A cosine similarity is dominated by whichever single line falls on the largest peak, so any phase with one reflection near a strong peak scores ≈ 1 .	Insufficient.
3. Coherent scaling + proportionality + minimum line count	Replace the cosine by a test that the implied scale is consistent <i>across</i> lines, and require a phase to have enough lines in range to be judged at all.	Improved, but on a real specimen still ranked quartz 56th of 205 , behind ilmenite, pyrite, cementite and faujasite — while quartz was demonstrably present, being the zero-error standard.	Insufficient.
4. Competitive non-negative least-squares screen	Ask the question that actually matters: does this phase explain intensity that nothing else can? Every candidate is fitted simultaneously with the clay library and with every other candidate, and ranked by the share of the pattern it takes.	On the same real specimen: quartz 7.4 %, albite 6.3 % , then a clean gap to 1.6 % for the next candidate. On a second specimen, albite 6.7 %, quartz 6.2 %, microcline 3.2 %. Adding the identified phases to the fit lowered R_{wp} .	Adopted.

Conclusion. Position matching cannot discriminate phases in a peak-rich pattern at these levels; explained intensity can. Method 4 is the default whenever a reference library is loaded, and method 3 is retained only as a fallback for when one is not. The residual tail at 1–2 % is least-squares slack rather than mineralogy, which is why the confirmation dialog leaves it unticked.

A.7 Crystallite thickness against measured peak width

Reasoning. The first fits to real specimens gave R_{wp} near 68 % while reproducing every peak *position* correctly, with the difference curve showing sharp excursions at the peaks — the signature of a width or position mismatch rather than a wrong phase. Crystallite thickness is the principal control on basal width (§2.6), so it was measured.

Method and results. Full widths at half maximum were measured on a real glycol-solvated mount and compared with the Scherrer prediction:

Position (°2 θ)	FWHM (°)	Assignment
8.82	0.117	illite 001
12.45	0.134	kaolinite 001 / chlorite 002
17.75	0.134	illite 002
26.57	0.134	quartz 101 / illite 003

A width of 0.117° at 8.82° implies a crystallite thickness of about 680 Å, or **68 layers**. The library in use had been built with a mean of 10 layers, which predicts a 001 width of **0.79°** — nearly seven times too broad.

Consequence. Crystallite thickness became a spanned library dimension (§3.3). Rebuilt with means of 5, 15 and 50 layers, the fit selected the $N = 50$ patterns for the sharp illite, confirming the mechanism.

An instructive negative result. Spanning thickness alone did *not* improve the agreement: R_{wp} went from 70.8 % to 70.8 %. The fit now had patterns of the right width available and chose them, but something else was still wrong — which led directly to the next two tests. Had the thickness test been taken as sufficient, the real causes would have been missed.

A.8 Layer spacing

Reasoning. With widths now correct, the residual remained concentrated at one place: the illite 001. At that point the measurement showed 7764 counts where the fit gave 1279 — the strongest clay reflection in the pattern almost entirely unexplained. A peak that is present in both but not fitted is a position problem.

Method and results. The basal spacing was determined from the measurement itself, using two orders so that the result is self-checking:

Order	Observed position (°2 θ)	d (Å)	Implied d(001) (Å)
001	8.884	9.9459	9.946
002	17.808	4.9768	9.954
003 (apparent)	26.631	3.3445	10.034 — <i>discard</i>

The 001 and 002 agree on $d(001) \approx 9.95$ Å. The third order appears to disagree, but that reflection is quartz 101 at 3.3435 Å overlapping illite 003, not illite alone — a reminder that the 3.34 Å peak of a clay pattern is rarely a clay peak only.

The library value, taken from the ICSD illite structure, is **10.022 Å**, placing the 001 at 8.816° against the observed 8.884°. The discrepancy of 0.066° is *half a peak width*. Scanning the spacing:

d(001) assumed (Å)	001 calculated (°)	Observed – calculated (°)
9.900	8.920	–0.038
9.950	8.886	–0.004
9.980	8.853	+0.029
10.000	8.836	+0.046
10.022 (library)	8.820	+0.062

Conclusion. The specimen's illite has $d(001) = 9.95$ Å, not the 10.022 Å of the reference structure. This is not an error in either: illite basal spacing varies from about 9.95 to 10.10 Å with interlayer potassium content and hydration, and a single refined structure is one point in that range. Layer spacing therefore became a second spanned library dimension. On rebuilding, the fit selected $d = 9.95$ Å of its own accord, and the illite 001 residual resolved.

A.9 The weighting of the fit

Reasoning. With widths and spacings now spanned, agreement was still poor. The remaining suspect was the fit's own weighting, which had been taken from the background-subtracted intensity. That is wrong on statistical grounds: the variance of a counting measurement is set by the number of photons recorded, and subtracting a smooth background does not reduce it. Weighting by the subtracted value gives near-zero-intensity points — the flat regions between peaks, which are pure noise — the largest weights, and the peaks the smallest.

Result. Changing the weights to $1/\max(\text{raw counts}, 1)$, and weighting the solve as well as the metric:

Weighting	R_{wp}	R_p
By background-subtracted intensity (incorrect)	70.8 %	48.2 %
By raw counts (counting statistics)	56.0 %	46.4 %

Conclusion. A 15-point improvement in R_{wp} from a change that alters no physics at all, only the statistical weighting. The lesson is general: when a background-subtracted pattern is fitted, the weights must still come from the counts that were actually measured.

A.10 Cumulative effect and the final state

Reasoning. The preceding tests each addressed one cause. Their cumulative effect on a single real specimen records what each was worth, and whether the final state is fit for use.

Stage	R_{wp}	Change made
Initial	68.4 %	—
+ thickness spanned	70.8 %	no improvement alone (§A.7)
+ correct weighting	56.0 %	§A.9
+ quartz in the fit	53.4 %	accompanying minerals matter
+ quartz, albite, calcite	49.1 %	—
+ layer spacing spanned	35.7 %	§A.8
+ minerals from the competitive screen	33.2 %	§A.6

Across specimens. Specimen 1, 33.2 %; Specimen 2, 34.7 %; Specimen 3, **76.1 %**. The third is a kaolinite-dominated separate that the program fits badly, and this is reported rather than concealed: its 7.17 and 7.37 Å pair is not matched by the kaolinite library, and the cause is not yet established. It is also one of the specimens in which quartz is absent, so its zero error had to be left unset.

Conclusion. Typical agreement on well-behaved specimens is $R_{wp} \approx 33\text{--}35\%$, which is reasonable for a fixed library that cannot refine positions or widths continuously, and poor by the standards of a Rietveld refinement, which can. At least one specimen type is fitted badly. The plots exist so that this is visible to the user in every case, and §9 states the limitation plainly.

A.11 End-to-end tests of the interface

Reasoning. Unit tests cannot show that the assembled program works: state may be lost between steps, a control may not be connected, a computation may be correct but never reach the screen. The complete workflow was therefore driven in a real browser, automatically, on real measurements, with the browser console monitored for errors.

Method. A script loaded three mounts, set the zero error on each, fitted and applied a background, loaded the phase database and reference library, ran the mineral search, confirmed the selection, ran both diagnostics, ran the fit, visited all four result plots, exported, and opened the simulator — capturing a screenshot at each step.

Results. The workflow completed with **no JavaScript errors**, all four plots rendering and all seven export files written. Two defects were found this way and would not have been found otherwise:

- **The confirmation dialog accepted itself.** Dash fires callbacks once for newly created components, so the dialog took whatever was ticked by default at the instant it appeared — the analyst never saw it, and three implausible minerals were silently added to the fit. The screenshot showed the selection already made. Fixed by guarding against the initial invocation.
- **Stale server state.** An early run reported results from before a change, because the development server does not reload edited code. Subsequent runs restart the server explicitly.

Conclusion. The assembled program works end to end on real data. The dialog defect in particular was invisible to unit testing, being a property of the interface framework rather than of any function, and was caught only by driving the interface and looking at what it displayed.

A.12 Summary

Area	Tests	Status
Interstratification model	25 automated, against closed-form and invariant results	Pass
Basal / three-dimensional consistency	9 automated, four phases	Pass
External validation	4 automated, against Reynolds (1965); $R = 0.039$ vs his 0.042	Pass
Workflow on synthetic data	7 automated	Pass
Structure import	205 phases; 12 positions to 0.03°; origin convention established	Pass
File readers	33 files; bit-for-bit against vendor export	Pass
Zero error	11 specimens $\times$ 3 mounts	Pass after correction
Mineral search	4 methods compared on known and real data	Method 4 adopted
Quantitative fit	3 real specimens	2 good, 1 poor and documented
Interface	End-to-end in a browser	Pass; 2 defects found and fixed

The automated suite comprises **45 tests** and runs in under two seconds (`pytest -q`). Ten of them are skipped automatically when structure data is not installed.

Appendix D — Provenance of the accompanying-mineral structures

The 209 structures below are those of the TOPAS structure library used during development. They are **not** distributed with ClayQuant and are not part of it: the library is user-supplied, imported by `clayquant-import-structures` (§4.3), and every ClayQuant installation may use a different one. The table is given so that any pattern the program calculates can be traced to the structure it came from, and that structure to its source database and entry.

Why the citations are not printed here. Of the 209 entries, 200 are ICSD, 3 AMCSD and 2 COD, and 4 carry no identifier. ICSD is a licensed database with no open lookup from an entry number to its publication, so those citations could not be retrieved and verified for this manual, and they have deliberately *not* been guessed: an invented reference is worse than an absent one. The identifiers below are exact, taken from the library file itself, and each resolves to its full bibliographic record in the database named — through an ICSD subscription, through rruff.geo.arizona.edu/AMS for AMCSD, or through crystallography.net for COD, all of which report the original publication for the entry. A

machine-readable copy with an empty `citation` column to fill in is in the repository as `docs/structure_provenance.csv`.

The four clay mineral structures are not in this table: they are integral to the program, are named in §4.2, and their publications are cited in full in §10.

Phase	Database	Entry
Actinolite	ICSD	68319
Aegirine	ICSD	156560
Akermanite	ICSD	158177
Albite	ICSD	68913
AliteM3	ICSD	94742
Almandine	ICSD	71895
AluminateC	ICSD	1841
AluminateO	ICSD	1880
Aluminium	ICSD	606001
Amorphous	—	—
Analcime	ICSD	156593
Anatase	ICSD	9852
Andalusite	ICSD	84613
Anglesite	ICSD	92609
Anhydrite	ICSD	16382
Ankerite	ICSD	66338
Anorthite	ICSD	34942
Apatite	ICSD	26204
Aphthitalite	ICSD	26018
Aragonite	ICSD	157992
Arcanite	ICSD	2827
Augite	ICSD	90143
Austenite	ICSD	186833
Azurite	ICSD	158577
Baricite	ICSD	202099
Barite	ICSD	16904
Bassanite	ICSD	79529
Bayerite	ICSD	26830
Belite Alpha H	ICSD	82997
Belite Beta	ICSD	81096
Biotite	ICSD	95359
Boehmite	ICSD	41215
Bornite	ICSD	1963

Bredigite	ICSD	9828
Brookite	ICSD	36408
Brownmillerite	ICSD	2842
Brucite	ICSD	79031
Brushite	ICSD	16132
Ca ₂ P ₂ O ₇	ICSD	14313
Calciolangbeinite	ICSD	290622
Calcite	ICSD	73446
Calcite-Superlorentz	ICSD	73446
Caledonite	ICSD	164674
Cassiterite	ICSD	262768
Celestine	ICSD	22322
Cementite	AMCSD	13523
Cerussite	ICSD	6178
Chalcopyrite	ICSD	2518
Chamosite	ICSD	80080
Chlorite	ICSD	164234
Chlorite Pawley	ICSD	164234
Chloroxiphite	ICSD	164137
Chromite	ICSD	171121
Chrysotile	ICSD	77747
Coesite	ICSD	100749
Copper	ICSD	53247
Corundum	ICSD	51687
Cotunnite	ICSD	43344
Covellite	ICSD	63327
Cristobalite	ICSD	75483
Cubanite	ICSD	42104
Cuprite	ICSD	63281
Diaboleite	ICSD	15382
Diamond	ICSD	52054
Diaspore	ICSD	29344
Dickite	ICSD	98132
Diopside	ICSD	17043

Dolomite	ICSD	100680
Dolomite-Ankerite SS	ICSD	100680
Dolomite-Superlorentz	ICSD	100680
Ellestadite	ICSD	262245
Enstatite	ICSD	9328
Enstatite-Ferrosilite SS	ICSD	9328
Epidote	ICSD	168464
Epsomite	ICSD	16595
Eskolaite	ICSD	75577
Ettringite	ICSD	155395
Faujasite	ICSD	40931
Fayalite	ICSD	54029
Ferrosilite	ICSD	34858
Fluorite	ICSD	60559
Forsterite	ICSD	9685
Forsterite-Fayalite SS	ICSD	9685
Franklinite	ICSD	91827
Galena	ICSD	63095
Gehlenite	ICSD	158171
Ghiaraite	AMCSD	19787
Gibbsite	ICSD	184708
Goethite	ICSD	245057
Goslarite	ICSD	107287
Graphite	ICSD	76767
Griceite	ICSD	41409
Grossular	ICSD	83461
Grunerite	ICSD	24214
Gypsum	ICSD	151692
Halite	ICSD	181148
Hematite	ICSD	40142
Hercynite	ICSD	40093
Heulandite	ICSD	37064
Hornblende	ICSD	158242
Hydrocalumite	ICSD	51890

Hydrocerussite	ICSD	280932
Hydrotalcite	ICSD	81963
Illite	ICSD	90144
Illite Pawley	ICSD	90144
Ilmenite	ICSD	30664
Iron	ICSD	76747
Jarosite	ICSD	166799
Jasmondite	ICSD	26407
K ₂ ZnCl ₄	—	—
Kaolinite 1M	ICSD	63192
Kaolinite 2M	—	—
Kieserite	ICSD	68345
LaB ₆	ICSD	248440
Labradorite	ICSD	100235
Langite	ICSD	30724
Laurionite	ICSD	28035
Lead	ICSD	96501
Lepidocrocite	ICSD	108876
Leucite	ICSD	161630
Lime	ICSD	163628
Litharge	COD	9012700
Lizardite	ICSD	202358
Maghemite	ICSD	79196
Magnesioferrite	ICSD	41290
Magnesite	ICSD	73731
Magnesite-Siderite	ICSD	73731
Magnetite	ICSD	158504
Malachite	ICSD	262794
Marcasite	ICSD	42415
Massicot	ICSD	15403
Mayenite	ICSD	241243
MCPM	ICSD	133
Melanterite	ICSD	157296
Mendipite	ICSD	93003

Merwinite	ICSD	26002
Microcline	ICSD	83531
Minium	ICSD	42484
Monetite	ICSD	2745
Montmorillonite	ICSD	51636
Mullite	ICSD	75305
Muscovite	ICSD	180081
Natron	ICSD	97922
Newberyite	ICSD	31281
Nordstrandite	ICSD	16018
Oldhamite	ICSD	30236
Orthoclase	AMCSD	20540
Otavite	ICSD	156740
Paralaurionite	ICSD	74291
PbS ₂ O ₃	ICSD	69532
PbSO ₃	ICSD	68889
Periclase	ICSD	159367
Perovskite	ICSD	62149
Phosgenite	ICSD	4240
Pigeonite-C	ICSD	9248
Pigeonite-P	ICSD	9247
Portlandite	ICSD	64950
Prehnite	ICSD	164238
Pseudoboleite	ICSD	67680
Pumpellyite	ICSD	158186
Pyrite	ICSD	15012
Pyrope	ICSD	71892
Quartz	ICSD	67121
Qusongite	ICSD	246149
Riebeckite	ICSD	200423
Rozenite	ICSD	23912
Rutile	ICSD	36412
Sanidine	ICSD	90141
Scotlandite	ICSD	30993

Siderite	ICSD	100678
Silicon	ICSD	51688
Sillimanite	ICSD	66330
Smithsonite	ICSD	100679
Spessartine	ICSD	83458
Sphalerite	ICSD	77090
Spinel	ICSD	77583
Spurrite	ICSD	156625
Staurolite	ICSD	53148
Stilbite	ICSD	83469
Stishovite	ICSD	40098
Sylvite	ICSD	165593
Syngenite	ICSD	157072
Taenite	ICSD	103555
Talc	ICSD	100682
TCP Alpha	ICSD	923
Tenite-Superstructure	ICSD	56386
Tenorite	ICSD	67850
Thaumasite	ICSD	185680
Titanite	ICSD	50144
Tobermorite11A	ICSD	92942
Tridymite	ICSD	34867
Troilite	ICSD	35005
Trona	ICSD	192710
Vaterite	ICSD	181959
Vermiculite	ICSD	192347
Vivianite	ICSD	423390
Wadalite	ICSD	151561
Weddellite	ICSD	192661
Whewellite	ICSD	153499
Whitlockite (Ca,Mg) ₃ (PO ₄) ₂	ICSD	1490
Whitlockite Ca ₃ (PO ₄) ₂	—	—
Wollastonite	ICSD	201538
Wuelfingite	ICSD	50447

Wuestite	ICSD	27237
Wurtzite	ICSD	67453
Yeelimite	COD	4001772
Zincite	ICSD	154486
Zinkosite	ICSD	71018
Zircon	ICSD	9582

Entries without an identifier are either not structural models (the amorphous and mixed-layer entries) or were added to the library without a database reference. Two entries of the library are Pawley (whole-pattern, structureless) phases for illite and chlorite; they carry the ICSD numbers of the structures their cells were taken from, which are the same illite and chlorite structures ClayQuant itself uses.

Appendix B — File formats read

Format	Extension	Notes
PANalytical XRDML	.xrdml	Intensities from <counts> or <intensities>; multiple appended scans are concatenated; positions from a start/end range or an explicit list. Wavelengths, goniometer radius, divergence slit, counting time and sample identifier are read as metadata.
Two-column text	.xy, .txt, .dat, .asc	2 θ and intensity. Comment and header lines are skipped; comma, tab and space separation are accepted.
Bruker RAW	.raw	Binary; the common RAW1.01 and RAW4 variants. Support is best-effort — if a file is not read, export it as two-column text from the vendor software.
CIF	.cif	The subset produced by ICSD, COD and AMCSD exports: cell, an explicit symmetry-operation loop, and an atom-site loop with occupancies and B_{iso} or U_{iso} .
TOPAS structure library	.xml	jEdit macro-menu form; str blocks with cell, space group and site lines.
Pattern library	.npz	Compressed NumPy archive holding the grid, the pattern matrix, and per-entry and library metadata.

Appendix C — Symbols

Symbol	Meaning	Units
s	scattering vector length, $2 \sin\theta/\lambda = 1/d$	$\text{\AA}^{-1}$
q	$\sin\theta/\lambda = s/2$; argument of scattering and displacement factors	$\text{\AA}^{-1}$
d	interplanar spacing; d_A , d_B layer repeat distances	$\text{\AA}$
$F(s)$	layer structure factor, complex	electrons
B	isotropic displacement parameter	$\text{\AA}^2$
N , $\langle N \rangle$	layers per crystallite; its mean over the CSDS	—
β	width of the lognormal CSDS in $\ln N$	—
$G(n)$	pair weight at layer separation n	—
$\mathbf{P}$, P_{mn}	Markov transition matrix and its elements	—
$\mathbf{W}$, W_m	stationary layer proportions	—
$\mathbf{T}$	phase-weighted transition matrix, equation (2.10)	—
R	Reichweite, the range of the stacking correlation	—
r	March–Dollase preferred-orientation parameter	—
α	angle between a scattering vector and the texture axis c^*	rad
γ	equatorial divergence of the incident beam	rad
$R_{\text{goniometer}}$	goniometer radius	mm
c_p	fitted scale factor of library pattern p	—
w_i	statistical weight of point i	counts $^{-1}$
R_{wp} , R_p	weighted-profile and profile agreement factors	—

ClayQuant 0.1.0 — user and reference manual. The program is distributed under the MIT licence. Structure data is not distributed with it and remains subject to the licence of its source. This manual documents the state of the program at the revision noted on the title page; the test record of Appendix A refers to that revision.